

1. Augmented State–Space Model Using a Pade Approximation

A. Pade Approximation of the Delay

For an input delay of length τ , the delay term is

$$e^{-\tau s}.$$

A second-order Pade approximation is given by

$$e^{-\tau s} \approx \frac{-\frac{\tau^2}{12}s^2 + \frac{\tau}{2}s - 1}{\frac{\tau^2}{12}s^2 + \frac{\tau}{2}s + 1}.$$

This rational function can be converted into a state–space realization with matrices (A_d, B_d, C_d, D_d) .

B. Augmenting the Plant

Assume the original plant is described by

$$\dot{x}(t) = A x(t) + B u(t), \quad y(t) = C x(t) + D u(t).$$

After replacing the delay $e^{-\tau s}$ with its Pade approximation, we introduce additional states x_d from the approximation. The augmented state vector is defined as

$$x_a = \begin{bmatrix} x \\ x_d \end{bmatrix}.$$

The augmented dynamics become

$$\dot{x}_a(t) = A_a x_a(t) + B_a u(t),$$

with

$$A_a = \begin{bmatrix} A & B C_d \\ 0 & A_d \end{bmatrix}, \quad B_a = \begin{bmatrix} B D_d \\ B_d \end{bmatrix}.$$

C. LQR Design on the Augmented Model

The cost function to be minimized is:

$$J = \int_0^\infty (x_a(t)^\top Q_a x_a(t) + u(t)^\top R u(t)) dt,$$

where a typical choice is to set

$$Q_a = \begin{bmatrix} Q & 0 \\ 0 & Q_d \end{bmatrix},$$

with larger weights in Q for the physical states and smaller weights in Q_d for the delay states.

The Algebraic Riccati Equation (ARE) for the augmented system is:

$$A_a^\top S + S A_a - S B_a R^{-1} B_a^\top S + Q_a = 0,$$

and the optimal state-feedback gain is obtained as

$$K = R^{-1} B_a^\top S.$$

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2. Smith Predictor Approach

A. Delay-Free State-Space Model and LQR Design

Assume the delayed plant can be written as

$$G(s) = G_p(s) e^{-\tau s},$$

where $G_p(s)$ is the delay-free part. The delay-free state-space model is

$$\dot{x}(t) = A x(t) + B u(t), \quad y(t) = C x(t) + D u(t).$$

An LQR controller for this model minimizes the cost

$$J = \int_0^\infty (x(t)^\top Q x(t) + u(t)^\top R u(t)) dt.$$

The optimal control law for the delay-free system is

$$u(t) = -K x(t),$$

with K computed from the ARE:

$$A^\top S + S A - S B R^{-1} B^\top S + Q = 0.$$

B. Predictor for the Delayed Output

Because the actual plant output is delayed, a predictor is used to estimate the future state. A common predictor is:

$$\hat{x}(t) = e^{A\tau} x(t) + \int_0^\tau e^{A(\tau-s)} B u(t-s) ds.$$

The predicted state $\hat{x}(t)$ approximates the state at time $t + \tau$. Then the control law is applied based on the prediction:

$$u(t) = -K \hat{x}(t).$$

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